Analysis of nontrivial data entails lengthy calculations, which should be performed by utilizing statistical software. The project asks each group to analyze the given data sets and obtain important results from data analysis through effective use of R/Python. At the same time, you may appreciate the role of linear regression models in scientific inquiry.

STUDY 1: SIMPLE LINEAR REGRESSION ANALYSIS (CHAPTERS 4, 16, AND 17)

An insurance company wants to relate the amount of fire damage in major residential fires to the distance between the burning house and the nearest fire station. The study is conducted in a large suburb of a major city; a sample of 15 recent fires in this suburb is selected. The amount of damage (in thousands of dollars) and the distance between the fire and the nearest fire station (in miles) are recorded for each fire. The data are saved in the file Fires.csv.

Investigate the association between the two variables by conducting a complete linear regression analysis of the data. Summarize your findings in a professional report.

STUDY 2: MULTIPLE LINEAR REGRESSION ANALYSIS (CHAPTERS 18 AND 19)

A consulting company develops multiple linear regression models for executive salaries of its client firms. The company has found that models that use the natural logarithm of salary as the response variable have better predictive power than those using salary as the response variable. For one firm, 10 potential predictor variables were measured in a sample of 100 executives. The data are saved in the file Salaries.csv. The variables are described in the following table.

Variable	*Description*	*Type*
y	Natural logarithm of salary	Quantitative
x_1	Experience (years)	Quantitative
x_2	Education (years)	Quantitative
x_3	Bonus eligibility (1 if yes, 0 if no)	Qualitative
x_4	Number of employees supervised	Quantitative
x_5	Corporate assets (millions of dollars)	Quantitative
x_6	Board member (1 if yes, 0 if no)	Qualitative
x_7	Age (years)	Quantitative
x_8	Company profits in the past 12 months (millions of dollars)	Quantitative
x_9	Has international responsibility (1 if yes, 0 if no)	Qualitative
x_{10}	Company's total sales in the past 12 months (millions of dollars)	Quantitative

Investigate the association between the response variable and the 10 potential predictor variables by conducting a complete linear regression analysis of the data. Summarize your findings in a professional report.

NOTES

- You are required to work collaboratively within the group but independently between groups. Doing otherwise risks a charge of cheating, which may result in a grade of zero.
  - — Your contribution to the project may be evaluated by other group members.
  - — Your grade will be computed as the group grade multiplied by the percentage of your contribution (for example, if you only complete 50% of what you are assigned by the group, the percentage of your contribution is 50%).

- No other statistical software but R/Python should be used.
- All numerical results should be obtained from computer outputs (i.e., no hand-calculations.)
- The report should be typed up on letter-size papers, double spaced and with a font size of at least 11 points.
  - There are no specific requirements about the format of the report; it's up to each group.
  - Include the relevant computer output in the report.
    - Submit R/Python codes in a separate file.
  - **Write down the group members who participate in the project and their respective percentages of contribution** (for example, if a member completes 50% of his/her assigned part, then his/her contribution is 50%).
- Submission policy:
  - Only one group member needs to submit the report on behalf of the entire group.
  - No individual submission unless you are the only one in the group who works on the project.
  - Multiple attempts of submission are allowed on Brightspace; however, only the most recent submission will be kept.
  - See the syllabus for policies on a late submission.

GRADING CRITERIA

- Clarity of report
- Completeness of analyses
- Correctness of results and interpretations
- Quality of graphs and tables (if any)